INCH-POUND

A-A-52176
30 May 1990

COMMERCIAL ITEM DESCRIPTION

Resistors, Variable, Non-Wire-Wound, Precision

The General Services Administration
has authorized the use of
this commercial item description.

1.0 Abstract. This commercial item description covers the general requirements for precision, non-wire-wound, variable resistors whose electrical output (in terms of percent of applied voltage) is linear with respect to the angular position of the operating shaft. These resistors are capable of full-load operation at maximum ambient temperatures of 70°C and are suitable for continuous operation, when properly derated, to maximum temperatures of 125°C, (see figure 2).

2.0 Part Number.

AA52176	A	1	2	A	B	103
	Shaft length (2.1)	Moisture Resistance (2.2)	Life Characteristic (2.3)	Function Conformity Tolerance Characteristic (2.4)	Output Smoothness Characteristic (2.5)	Resistance (2.6)

2.1 Shaft length. The shaft length is identified by a single letter in accordance with table I.

2.2 Moisture resistance. Moisture resistance is identified by a single digit in accordance with table II.

2.3 Life characteristic. The life characteristic is identified by a single letter in accordance with table III.

FSC 5905

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2.4 Function-conformity tolerance characteristic. The function-conformity tolerance characteristic is identified by a single letter in accordance with table IV.

2.5 Output-smoothness characteristic. The output-smoothness characteristic is identified by a single letter in accordance with table V.

2.6 Resistance. The nominal total resistance value expressed in ohms is identified by a three-digit number. The first two digits represent significant figures and the last digit specifies the number of zeros to follow. When fractional values of an ohm, or values of less than 10 ohms, are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits of the group represent significant figures, as shown in the following example:

3R0 signifies 3.0 ohms
102 signifies 1000 ohms.

2.7 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3. Salient characteristics.

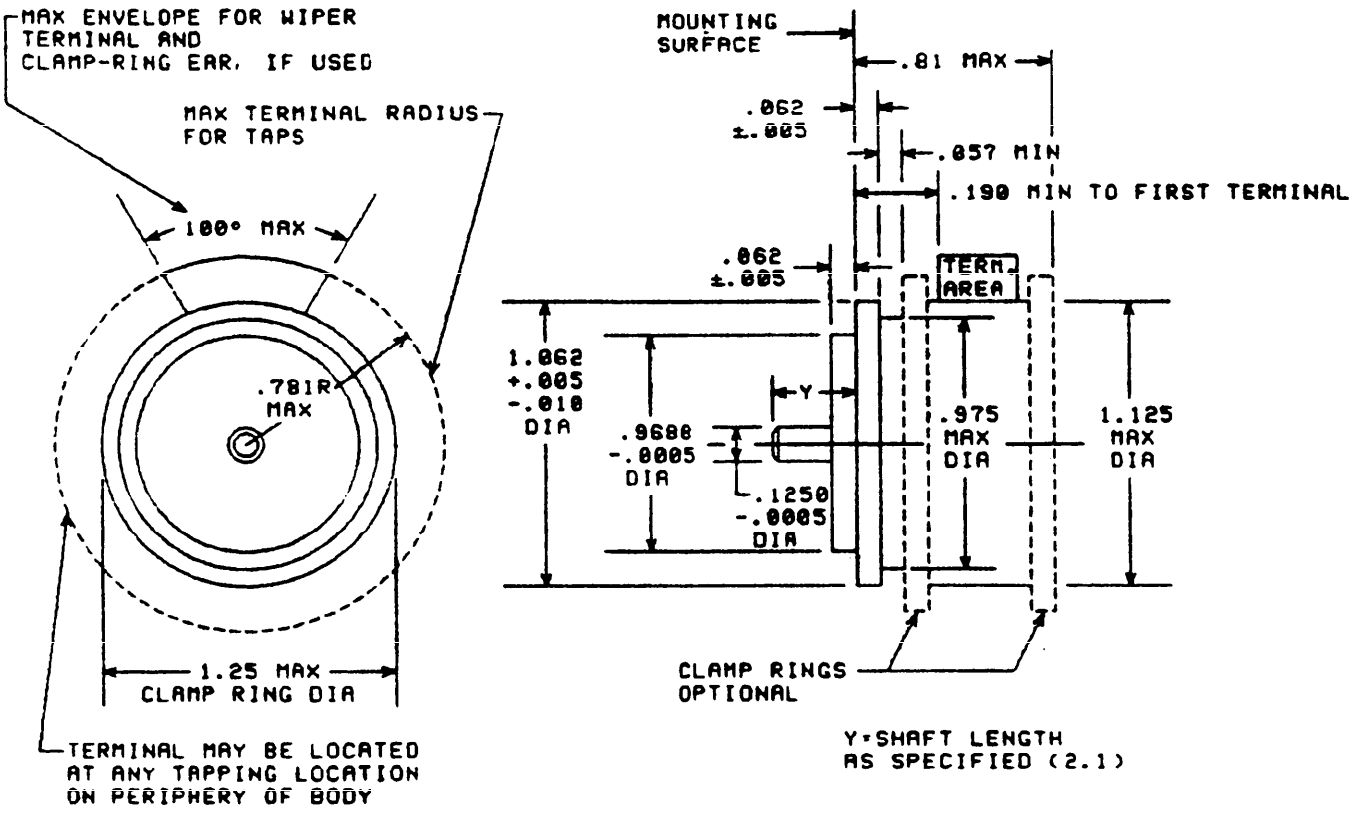
3.0 Maximum continuous working voltage. The maximum continuous working voltage for these resistors shall be 250 volts.

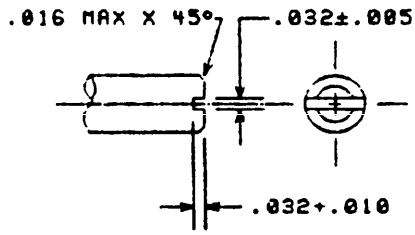
3.1 Power rating. The power rating for these resistors shall be 1.25 watts

3.2 Resistance-temperature characteristic and maximum ambient temperature. The ambient temperature with rated load is 70°C and the ambient temperature with zero load is 125°C. The resistance-temperature characteristic shall be in accordance with table IX.

3.3 Plastic. Plastic laminates containing a cotton-fabric base, or plastic-molding compounds containing a cotton or wood-flour filler shall not be used. Where not machined, plastic material shall have the original, smooth, or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable.

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SHAFT DETAIL

INCHES	MM	INCHES	MM
.0005	0.013	.62	15.8
.005	0.13	.781	19.84
.010	0.25	.81	20.6
.016	0.41	.9688	24.608
.032	0.81	.975	24.77
.057	1.45	1.062	26.97
.062	1.57	1.125	28.58
.1250	3.18	1.25	31.8
.190	4.83		

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS ARE GIVEN FOR GENERAL INFORMATION ONLY.

FIGURE 1. A-A-52176 resistor.

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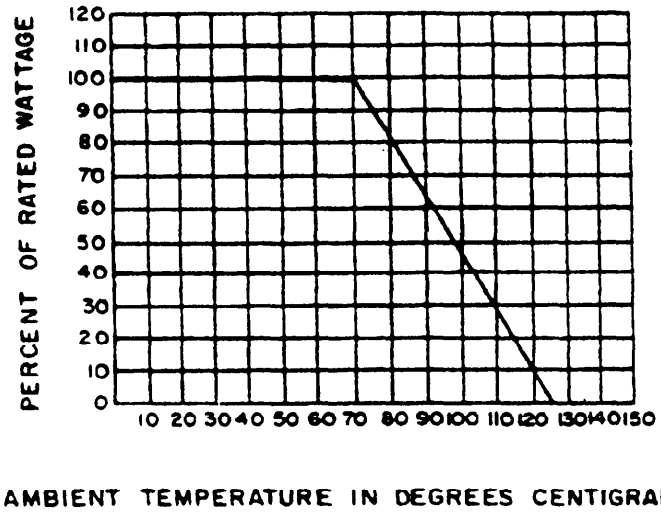


FIGURE 2. Derating curve for high ambient temperatures.

Table I. Shaft Length.

Symbol	Shaft Length ± 1/32 inch (.79mm)	
	Inch	mm
A	3/8	(9.53)
B	1/2	(12.70)
C	5/8	(15.88)
D	3/4	(19.05)

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Table II. Moisture resistance characteristic.

Symbol	Moisture Resistance (Total ΔR)
	Percent
1 ----------	± 5
2 ----------	± 10

Table III. Life characteristic.

Symbol	Life Total number of revolutions	Dither Life (In hours)
1	2.5×10^5	0.5
2	1×10^6	2
3	5×10^6	10
4	25×10^6	50

Table IV. Function-conformity tolerance characteristic.

Symbol	Function-conformity tolerance	
	Percent (±)	
	Initial	Degraded
A ------	1.0	1.5
B ------	0.5	0.75
C ------	0.25	0.375
D ------	0.10	0.15

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qualification of the required sample in ord

TABLE V. Output-smoothness characteristic.

Symbol	Peak-to-peak voltage (E_o/E_{in})	
	Percent	
	Initial	Degraded
A --------	2.0	2.2
B --------	0.5	0.7
C --------	0.1	0.15
D --------	0.025	0.04
E --------	0.01	0.02

Table VI. Nominal total resistance values and maximum end voltage.

Nominal total resistance values	Maximum end voltage (percent of total applied voltage)
Ohms	Percent
100	3.5
200	2.5
500	1.9
1,000	1.8
2,000	1.7
5,000	1.6
10,000	1.6
20,000	1.6
50,000	1.6
Megohms	
0.1	1.6
0.2	1.6
0.5	1.6
1.0	1.6

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3.4 Fungus. All external materials shall be nonnutrient to fungus growth or shall be suitably treated to retard fungus growth.

3.5 Ferrous metals. The use of ferrous metals, with the exception of stainless steel, is prohibited.

3.6 Operating shaft. Operating shafts shall be made from corrosion-resistant steel. The shafts shall be round and slotted and the length shall be in accordance with Table I.

3.7 Contact arm assembly. Contact pressure on the resistance element shall be maintained uniform by positive pressure and shall permit smooth electrical and mechanical control of the variable resistor over the entire range of continuity travel. The moving contact shall have continuous electrical contact with its terminal throughout the entire mechanical travel and shall be insulated from the operating shaft and case.

3.8 Terminals. The terminals shall be of rigid metal and shall be located as shown in figure 1. The terminals shall be suitably treated to facilitate soldering. Terminals shall be numbered with No.2 as the contact arm terminal, No.1 as the counterclockwise (CCW) terminal, and No.3 as the clockwise (CW) terminal and shall be located as specified.

3.9 Enclosure. Resistors shall have suitable enclosures for protection against mechanical damage.

3.10 DC Resistance. The total direct-current (DC) resistance shall not deviate from the specified nominal resistance value by more than plus or minus 10 (ten) percent.

3.11 Lateral runout. The lateral runout shall not exceed 0.002 inch-per-inch of mounting-surface radius or 0.001 inch, whichever is greater.

3.12 Shaft runout. The shaft runout shall not exceed 0.002 inch-per-inch of shaft length from mounting surface to point of measurement or 0.001 inch total, whichever is greater.

3.13 Pilot-diameter runout. The pilot-diameter runout shall not exceed 0.001 inch.

3.14 Shaft radial play. The radial play shall not exceed 0.001 inch.

3.15 Shaft and play. The end play shall not exceed 0.005 inch.

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3.16 Starting torque. The maximum value of the minimum torque required to initiate shaft rotation shall be 0.5 ounce-inch. The maximum value of the starting torque shall not increase to more than 150 percent of the initial value.

3.17 Running torque. The maximum value of the minimum torque to sustain shaft rotation shall be 0.4 ounce-inch. The maximum value of running torque shall not increase to more than 150 percent of the initial value.

3.18 Insulation resistance. The initial insulation resistance of the resistors shall not be less than 1,000 megohms. After degradation, the insulation resistance shall not be less than 500 megohms.

3.19 Dielectric withstanding voltage. The resistors shall be capable of withstanding a test voltage of 750 volts rms for resistors of 1.125-inch diameter and smaller or 1,000 volts rms for resistors having diameters larger than 1.125 and exhibit no damage, arcing, or breakdown. The leakage current shall not exceed 1 milliampere (MA).

3.20 Function-conformity. The function-conformity shall be independent linearity.

3.21 Function-conformity tolerance. The function-conformity tolerance shall be the value specified in Table IV. The resistors shall be capable of not exceeding the specified degradation value of Table IV for function-conformity tolerance after being subjected to a previous series of environmental or mechanical tests.

3.22 Mechanical backlash. The operating shaft shall not move in excess of the applicable value specified in Table VII without moving the contact arm.

TABLE VII. Mechanical backlash.

Function-conformity tolerance	Mechanical backlash
(Percent)	(°)
1.0	0.5
0.5	0.25
0.25	0.1
0.10	0.1

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3.23 Minimum voltage. The resistor shall be capable of not exceeding the minimum voltage expressed as a percent of applied voltage specified in Table VIII.

TABLE VIII. Minimum voltage.

Total resistance	Standard minimum percent of total applied oltage
Above 2 kilohms	0.1 percent maximum
Above 1 kilohm to 2 kilohms	0.2 percent maximum
Above 500 ohms to 1 kilohm	0.4 percent maximum
Above 200 ohms to 500 ohms	1 percent maximum
100 ohms to 200 ohms	2 percent maximum

3.24 End voltage. The maximum end voltage shall be as specified in Table VI.

3.25 Output smoothness. The initial output smoothness shall not exceed the initial values specified in Table V. The output smoothness shall not exceed the degradation values specified in Table V after exposure to environmental tests.

3.26 Terminal strength. The resistors shall be capable of submission to a terminal strength test without evidence of mechanical or electrical damage.

3.27 Thermal shock. The resistors shall be capable of withstanding a thermal shock test as described in method 107 of MIL-STD-202 and exhibit no evidence of mechanical damage, or opening of the element, and the total resistance change shall not exceed 10 percent of the initial resistance value.

3.28 Resistance-temperature characteristic. The resistance-temperature characteristic, referred to an ambient temperature of 25°C, shall not exceed the value specified in Table IX.

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TABLE IX. Resistance change from reference temperature.

Temperature	Maximum resistance change from reference temperature
°C	Percent
25	Reference
0	±2.5
-15	±4
-65	±5
25	Reference
50	±2.5
75	±5
105	±5
125	±5

3.29 Rotational load life. The resistor shall be capable of meeting the rotational life characteristic specified in Table III at a speed of 400 rpms.

3.30 Dither life. The resistor shall be capable of meeting the dither life characteristic specified in Table III.

3.31 Low temperature operation. The resistor shall be capable of operation for 4 hours at -65°C with rated voltage applied for 45 minutes with the mechanical and electrical connections not being adversely affected. Under these conditions, the resistor shall be capable of a rotational torque not exceeding two times the specified torque and a total resistance change not exceeding 5 percent of the initial resistance value.

3.32 Low-temperature exposure. The resistor shall be capable of twenty-fours exposure to a temperature of -65°C, stabilization at room temperature for twenty-fours and shall show no evidence of mechanical damage, opening of the element or a change in output ratio exceeding the degraded function-conformity tolerance specified in Table IV or 0.5 percent, whichever is greater.

3.33 High-temperature exposure. The resistors shall be capable of exposure to a temperature of 125°C for a period of 1000 hours and exhibit no evidence of mechanical damage, opening of the element or change in output ratio exceeding the degraded function-conformity tolerance specified in Table IV or 0.5 percent, whichever is greater.

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3.34 Shock (specified pulse). The resistors shall be capable of exposure to the shock (specified pulse) test of method 213 of MIL-STD-202, test condition I and exhibit no evidence of mechanical or electrical damage, nor momentary discontinuity in excess of 0.1 milliseconds (ms).

3.35 Vibration, high frequency. The resistors shall be capable of exposure to vibration, high frequency test of method 204 of MIL-STD-202, test condition B and exhibit no evidence of mechanical or electrical damage, momentary discontinuity in excess of 0.1 ms, nor a total resistance change exceeding 2 percent of the initial resistance value.

3.36 Salt spray (corrosion). The resistors shall be capable of exposure to the salt spray corrosion test of method 101 of MIL-STD-202, test condition A, and exhibit no appreciable corrosion when the units are disassembled and inspected.

3.37 Moisture resistance. The resistors shall be capable of exposure to the moisture resistance test of method 106 of MIL-STD-202 with insulation resistance not less than 10 megohms, or 100 times the resistance value, whichever is greater, and the total resistance change shall not exceed the values of Table II.

3.38 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance.

4.0 Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or purchase order.

5.0 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Preservation, packaging, packing, labeling and marking requirements.

MILITARY INTERESTS:

Custodians:

Army - ER

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-7FXE

PREPARING ACTIVITY:

Army - ER

(Project 5905-1222)

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